MIL-F-45465F (MU)
28 AUGUST 1973
SUPERSEDING
MIL-F-45465E (MU)
11 SEPTEMBER 1970

MILITARY SPECIFICATION

FUZE, PD, M524A6
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing of one type of fuze designated as Fuze, PD, M524A6.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of
MIL-D-48116 - Delay Element, M2
MIL-P-60412 - Packing, Packing and Marking for Shipment of Artillery Type and Rocket Fuzes, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for

FSC: 1390

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MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

9205729 - Fuze, PD, M524A6, Assembly

PUBLICATIONS

ORDNANCE CORPS

ET-9205729 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. - Material and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly. - The assembly shall comply with all requirements specified on Drawing (dwg.) 9205729 and with all requirements specified in applicable specifications.

3.3 Jolt and Jumble. - The assembly shall comply with the requirements of Test No. 101 and 102 of MIL-STD-331.

3.4 Transportation Vibration. - The assemblies shall comply with the requirements of Test No. 104 of MIL-STD-331.

3.5 Functioning. - The fuze shall cause the shell to function high order when tested as specified in 4.2.3.4.

3.6 Premature Burst. - The fuze shall show no evidence of a premature burst when tested as specified in 4.2.3.4. A premature burst is defined as the functioning of the assembly in the mortar tube or in flight before impact.

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3.7 Arming Delay. - The fuze shall not function on impact against a plywood screen when tested as specified in 4.2.3.4.

3.8 Workmanship. - All parts shall be free of chips, dirt, grease, rust and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to defined terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product.- At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 Inspection Provisions

4.2.1 Lot Formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Parts and assemblies of one lot interfix number from one manufacturer.
b. Detonator assemblies of one lot interfix number from one manufacturer.
c. Delay elements of one lot interfix number from one manufacturer.
d. Plunger lead assemblies of one lot interfix number from one manufacturer.
e. Booster lead assemblies of one lot interfix number from one manufacturer.
f. Booster pellets of one lot interfix number from one manufacturer.

4.2.2 Examination. - The examinations listed in the following sub-paragraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an

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AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable sub-paragraph of this section, except where 100 percent is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Pellet, booster (see dwg. 8797518 covering a detail of dwg. 9205729)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 2.50 percent		
201.	Weight	Balance	01001
202.	Diameter	Gage	01002
203.	Length	Gage	01003
204.	Cracks, chips or broken	Visual	01004
205.	Evidence of poor workmanship (see 3.8)	Visual	01005

4.2.2.2 Booster lead assembly (see dwg. 8797521 covering a detail of dwg. 9205729).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Total length, max.	Gage	02001
102.	Length of lower cup, max. ...	Gage	02002
Minor:	AQL 1.0 percent		
201.	Total length, min	Gage	02003
202.	Crimp not full 360 degrees	Visual	02004
203.	Evidence of poor workmanship (see 3.8)	Visual	02005

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4.2.2.3 Arming mechanism loading assembly (see dwg. 8797530 covering a detail of dwg. 9205729).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Detonator cap torque	Gage	03001
102.	Detonator inverted (red end up)	Visual	03002
103.	Detonator cap stake missing .	Visual	03003
104.	Cushion missing	Visual	03004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	03005

4.2.2.4 Fuze, PD, M524A6 Assembly, prior to assembly of Booster Cup (see dwg. 9205729).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Arming mechanism and plunger assembly not fully seated	Gage	04001
102.	Booster pad missing	Visual	04002
103.	More than one booster pad	Visual	04003
104.	Booster lead assembly missing.	Visual	04004
105.	Booster assembly missing or not inverted	Visual	04005
106.	Plunger assembly missing	Visual	04006
107.	Foreign material inside assembly after setback pin has been installed	Visual	04007
Minor:	AQL 1.50 percent		
201.	Length of booster assembly ...	Gage	04008
202.	Booster assembly disc punctured	Visual	04009

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203.	Booster lead tape missing	Visual	04010
204.	Evidence of poor workmanship (see 3.8)	Visual	04011

4.2.2.5 Fuze, PD, M524A6 Assembly (see dwg. 9205729).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Pull wire assembly missing	Visual	05001
2.	Warning label missing	Visual	05002
Major:	AQL 0.65 percent		
101.	Length from flange to bottom of booster cup, max.	Gage	05003
102.	Booster cup torque	Gage	05004
103.	Threads damaged	Visual/ Gage	05005
Minor:	AQL 1.5 percent		
201.	Fuze setting incorrect	Visual	05006
202.	Marking missing, incorrect or indistinct	Visual	05007
203.	Evidence of poor workmanship (see 3.8)	Visual	05008

4.2.2.6 Body and striker assembly (see dwg. 9205730 covering a detail of dwg. 9205729).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Foreign material inside assembly after plunger pin of pull wire has been installed	Visual	06001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	06002

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4.2.2.7 Arming mechanism and plunger assembly (see dwg. 9205732 covering a detail of dwg. 9205729).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Arming mechanism and plunger assembly in armed position	Visual	07001
Minor:	AQL 0.65 percent		
201.	Length, max.	Gage	07002
203.	Evidence of poor workmanship (see 3.8)	Visual	07003

4.2.2.8 Plunger assembly (see dwg. 9205737 covering a detail of dwg. 9205729).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Depth to top of delay element	Gage	08001
102.	Detonator inverted	Visual	08002
103.	Any component or operation missing	Visual	08003
104.	Flash hole obstructed	Visual	08004
105.	Delay element stake missing ...	Visual	08005
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	08006

4.2.2.9 O ring (see dwg. 9234584 covering a detail of dwg. 8797530).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 percent		
101.	Inside diameter, min	Gage	09001
102.	Outside diameter, max.	Gage	09002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	09003

4.2.3 Testing

4.2.3.1 Jolt and Jumble (see 3.3) - Critical Defect - Code No. 05009. - This test shall be performed as specified in 4.3.1 using equipment in accordance with 4.2.4. As a minimum, inspection for Critical Defects shall include:

a. The assembly shall not be broken or become disassembled to any degree (exclusive of broken creep pin or rotor drive lug).

b. The pull wire assembly shall not become disengaged from the fuze or be broken.

c. The pallets shall not be bent, cracked, broken or missing from the balance assembly.

d. The plunger shall not be cracked or broken.

e. The striker guide pin shall not have moved past the lock stake.

4.2.3.1.1 Beginning with the first lot produced and continuing until three consecutive lots from each metal parts contractor have complied with the acceptance criteria specified in the applicable Military Standards, thirty-two (32) fuze assemblies shall be selected from each lot for these tests. If one or more fuze assemblies fail to comply with the requirement specified, the lot shall be rejected.

4.2.3.1.2 After three consecutive lots have complied with 4.2.3.1.1, the sample size shall be reduced to twelve (12) for all successive lots. If one or more fuze assembly fails to comply with the requirement specified, the lot shall be rejected.

4.2.3.2 Transportation Vibration Test (see 3.4) - Critical Defect for Safety Deficiencies - Code No. 05010; Major Defect for Operation Deficiencies - Code No. 05011. - This test shall be performed as specified in 4.3.2 using equipment in accordance with 4.2.4. As a minimum, inspection for Critical Defects shall include:

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a. The assembly shall not be broken or become disassembled to any degree (exclusive of broken creep pin or rotor drive lug).

b. The pull wire assembly shall not become disengaged from the fuze or be broken.

c. The pallets shall not be bent, cracked, broken or missing from the balance assembly.

d. The plunger shall not be cracked or broken.

e. The striker guide pin shall not have moved past the lock stake.

As a minimum, inspection shall be made for the following Major Defects:

a. The retaining plate shall not be loose.

b. The setback screw shall not be loose or disengaged.

c. The striker plug shall not be loose or disengaged from the striker.

d. The firing pin shall not be disengaged, either partially or fully, from the striker.

4.2.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots from each metal parts contractor have complied with the acceptance criteria specified in the applicable Military Standard, thirty-two (32) fuze assemblies shall be selected from each lot for this test. If one or more fuze assemblies fail the operational or safety requirement, the lot shall be rejected.

4.2.3.2.2 After three consectuve lots have complied with 4.2.3.2.1, the sample size shall be reduced to ten (10) for all successive lots. If one or more fuze assemblies fail the operational or safety requirement, the lot shall be rejected.

4.2.3.3 Functioning (Proving Ground) - Code No. 05012. - The fuzes shall be observed for compliance with the requirements classified in Table I.

TABLE I

DEFECT	DEFECT CLASSIFICATION
(a) Premature Burst (see 3.6)	Critical
(b) Failure to Function High Order (see 3.5)	Major

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4.2.3.3.1 Functioning. - One hundred fifty (150) fuzes shall be randomly selected from each lot and assembled to the appropriate round at a Government proving ground and fired according to instructions in 4.3.3. If five or more of the fuze assemblies fail to function the shell high order, or if a premature burst occurs at any time, the lot shall be rejected.

4.2.3.4 Arming Delay - Critical Defect - Code No. 05013 (see 3.7)

4.2.3.4.1 Twelve (12) fuze assemblies shall be randomly selected from each lot for the purpose of ballistically testing for arming delay in accordance with 4.3.4. If one or more of the fuzes fail to meet the requirement of 3.7, the lot shall be rejected.

4.2.3.5 Check Test for Possible Deterioration of Delay Element, M2 (see Specification MIL-D-48116) - Major Defect - Code No. 10001. - If the total elapsed time between the original acceptance of the M2 delay lot and the assembly of the lot into fuzes exceeds two years, or if the elements have been subjected to adverse conditions, however brief, at any time since previous tests, the delay element lot shall be subjected to and must satisfactorily pass the functioning test specified in the applicable specification immediately before the delay lot is assembled into the fuzes. This test shall be performed by the contractor on elements selected by the Government inspector at the facility assembling and loading the fuzes (see 6.3).

4.2.4 Inspection Equipment. - The inspection equipment required to perform the examinations and tests prescribed in this section is identified by Equipment Tabulation ET-9205729 (see 6.2).

4.2.4.1 Government Rights to Documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Government Use of Contractor's Inspection and Test Equipment. - The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

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4.3 Test Methods and Procedures

4.3.1 Jolt and Jumble. - The loaded fuze assemblies shall be tested in accordance with the procedure specified in Standard MIL-STD-331 except that after completion of the jolt test, the assemblies shall not be disassembled but shall be subjected to the jumble test, utilizing a jumble fixture as specified in 4.2.4, after which the samples shall be disassembled to determine compliance with the requirements.

4.3.2 Transportation Vibration. - The loaded fuze assemblies shall be tested in accordance with Standard MIL-STD-331 for compliance with the requirements.

4.3.3 Functioning (Proving Ground). - The rounds shall be fired from a 81MM mortar at between 45° and 75° elevation. One half of the sample rounds shall be fired at 0 propellant increment phase and the balance fired at 9 propellant increment phase with 40 assemblies set at Superquick and 35 set Delay in each increment.

4.3.4 Arming Delay (Proving Ground). - The fuze assemblies shall be assembled to inert filled, appropriate projectiles, set Superquick and fired at Charge Zone 0 from an 81MM mortar. They shall be fired against a 1/4 inch screen set at a distance of 200 feet plus or minus one inch. (The muzzle velocity of each round shall be reported in the official firing record).

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Packaging, packing, marking and inspection thereof shall be in accordance with the requirements of Specification MIL-P-60412 for the level specified.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

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6.2 Inspection Equipment Designs. - Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers on the Equipment Lists referenced on the Equipment Tabulation. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE Designs. - SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with SIE designs if not otherwise covered thereby (see 6.2.1.c). Designs shall be of the category and form (per MIL-D-1000 specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

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6.2.3 Submission of Designs for Approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-F, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 Cost of Deterioration Check Test. - The Contracting Officer will arrange to have the contractor reimbursed for costs incurred in the performance of the check test (see 4.2.3.5).

6.4 Proving Ground Test Requirements:

	Sample Size				Acceptance Criteria			
	Charge				Functioning Failures		Premature Burst	
Test	0		9		Accept	Reject	Accept	Reject
	SQ	D	SQ	D				
Functioning	40	35	40	35	4	5	0	1
Arming Delay	12	-	-	-	-	-	0	1

6.5 Submission of Data. - In addition to the normal distribution of records, when the fuze is procured by the Department of the Army, one copy of the ammunition data card in accordance with MIL-STD-1167, and one copy of all ballistic test data will be forwarded to each of the following:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-F
SARPA-AD-F-E
Dover, New Jersey 07801

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1390-A-037
14

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